Fundamentals of Database System

Chapter: 2

Review Questions: 2.1

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Define the following terms:

1) Data model
2) Database schema
3) Database state
4) Internal schema
5) Conceptual schema
6) External schema
7) Data independence
8) DDL
9) DML
10) SDL
11) VDL
12) Query language
13) Host language
14) Data sublanguage
15) Database utility
16) Catalog
17) Client/server architecture

18) Three- tier architecture

19) N- tier architecture

1) What is Data Model?

A data model is a set of concepts used to describe the structure of a database, including data types, relationships, constraints, and basic operations.

2) What is Database Schema?

The description of a database is called the database schema or a database schema is the overall design of the database (tables, fields, relationships), which is specified during database design and is not expected to change frequently.

3) What is Database State?

The data in the database at a particular moment in time is called a database state or snapshot, and it also called the current set of occurrences or instances in the database.

4) What is Internal Schema?

The internal level has an internal schema, which describes the physical storage structure of the database or describe how data is physically stored in the system. The internal schema uses a physical data model and describes the complete details of data storage and access paths for the database.

5) What is Conceptual Schema?

Describes the overall structure of the database for the whole system. The conceptual level has a conceptual schema, which describe the structure of the whole database for a community of users. The conceptual schema hide the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations and constraints. Usually, a representational data model is used to describe the conceptual schema.

Often based on conceptual schema design in a high-level data model.

6) What is External Schema?

Shows how different users view the data. The external or view level includes a number of external schema or user view. Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group. It is also based in a high- level conceptual data model.

7) What is Data independence?

The ability to change database structure without affecting applications. The three- architecture can be used to further explain the concept of data independence, which can be defined as the capacity to change the schema at one level of a database system without having to change the schema at the next high- level. We can define the two types of data independence:

1. Logical data independence 2. Physical data independence

8) What is DDL (Data Definition Language)?

Is used by the DBA and by database designers to define both schemas or used to define database structure(create, alter, drop). The DBMS will have a DDL compiler whose function is to process DDL statements in order to identify descriptions of the schema constructs and to store the schema description in the DBMAS catalog. The DDL is used to specify the conceptual schema only.

9) What SDL (Storage Definition Language)?

Is used to specify the internal schema or defines how data is stored internally. The mappings between the two schemas may be specified in either one of these languages. In most relational DBMSs today, there is no specific language that performs the role of SDL. Instead, the internal schema is specified by a combination of functions, parameters, and specifications related to storage of files. These permit the DBA staff to control indexing choices and mapping of data storage.

10) What is DML (Data Manipulation Language)?

Once the database schemas are compiled and the database is populated with data, users must have some means to manipulate the database. Typical manipulations include retrieval, insertion, deletion, and modification of the data. The DBMS provides a set of operations or a language called the data manipulation language (DML) for these purposes. There are two main types of DMLs, 1. High- level or nonprocedural DML 2. Low- level or procedural DML.

11) What is VDL (View Definition Language)?

To specify user views and their mappings to the conceptual schema, but in most DBMSs the DDL is used to define both conceptual and external schemas. In relational DBMSs, SQL is used in the role of VDL to define user or application views as results of predefined queries.

12) What is Host Language and Data Sublanguage?

Whenever DML commands, whether high level or low level, are embedded in a general- purpose programming language, that language is called the host language and the DML is called the data sublanguage.

13) What is Query Language?

A high level DML used in a standalone interactive manner is called a query language. In general, both retrieval and update commands of a high level DMAL may be used interactively and are hence considered part of the query language. Casual end users typically use a high level query language to specify their requests, whereas programmers use the DML in its embedded form. For naive and parametric users, there usually are user- friendly interfaces for interacting with the database. These can also be used by casual users or others who do not want to learn the details of a high- level query language.

14) What is Database utility?

Database utility help the DBA to manage the database system or tools used for maintenance. Common utilities have the following types of functions:

- Loading
- Backup
- Database storage reorganization
- Performance monitoring

15) What is Catalog?

A collection of metadata (data about data).

16) What is Client/Server Architectures?

The client/server architecture was developed to deal with computing environments in which a large number of PCs, workstations, file servers. Printers, database server, web servers, e- mail servers, and other software and equipment are connected via a network.

17) What is Three- tier Architecture?

The three- tier architecture, which adds an intermediate layer between the client and the database server. This intermediate layer or middle tier is called the application server or the web server, depending on the application.

18) What is N- tier Architecture?

An architecture with multiple layers beyond three tiers.

References:

The content of this project is based on standard Database Management System (DBMS) concepts and is supported by reliable online learning resources.

The following sources were used for understand and simplifying the topics:

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- https://www.geeksforgeeks.org/dbms/

2. TutorialsPoint- DBMS Tutorial

- https://www.tutorialspoint.com/dbms/index.htm

3. Javatpoint- DBMS Tutorial

- https://www.javatpoint.com/dbms- tutorial

4. W3Schools- SQL and Database Concepts

- https://www.w3schools.com//sql/

5. IBM Document- Database Concepts and Architecture

- https://www.ibm.com/docs/en

These references helped in explaining core concepts such as data model, database schemas, and database architecture in a clear and structured way.

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